WIGVO: Real-Time Bidirectional Speech Translation over Legacy PSTN Calls via Dual-Session Echo Gating

Hyeongseob Kim
SoundMind Inc.
rukais2294@gmail.com

Sangwoo Son
SoundMind Inc.
xhxh9539@hs.ac.kr

Jinmo Kim
SoundMind Inc.
morirokim@gmail.com

Hyeonsang Kim
SoundMind Inc.
hyeonsangyy@gmail.com

Hyunwoo Cho
SoundMind Inc.
gus007dn@gmail.com

Abstract

Real-time speech translation with large language models (LLMs) has become feasible in controlled settings such as mobile apps and web browsers, where developers assume wideband audio and client-side echo cancellation. Recent end-to-end full-duplex models further push latency below 200 ms but remain confined to broadband environments. However, deploying such systems over the Public Switched Telephone Network (PSTN) remains challenging due to narrowband G.711 audio, unpredictable round-trip delays, and absence of client-side signal processing. We present **WIGVO**, a server-side relay system that enables bidirectional LLM-based speech translation over ordinary telephone calls without requiring app installation or carrier integration. A central contribution is addressing what we term *echo-induced self-reinforcing translation loops*: synthesized speech echoing back through the PSTN gets re-ingested and repeatedly translated. WIGVO solves this through a dual-session architecture with deterministic silence injection and energy-based voice activity detection (VAD) gating. We evaluate WIGVO on 96 completed PSTN calls (37 fully instrumented) across three communication modes—voice-to-voice (V2V), text-to-voice (T2V), and full-agent—observing 562 ms median caller-to-callee latency and 1,994 ms median callee-to-caller latency, zero echo-induced translation loops, and USD 0.42 per minute cost. The system is deployed at https://wigvo-web-gzjzn35jyq-du.a.run.app with a live demo and video walkthrough.

1 Introduction

Streaming speech-to-speech translation has advanced rapidly (Seamless Communication et al., 2023a; OpenAI, 2024), but existing systems assume wideband audio and client-side acoustic echo cancellation (AEC) (ITU-T, 2015). The Public Switched Telephone Network (PSTN) remains the primary inbound interface for local businesses, hospitals, and government offices (International Telecommunication Union, 2020), yet encodes audio via G.711 μ-law at 8 kHz (ITU-T, 1988), introduces 80–600 ms delays (ITU-T, 2003), and provides no AEC. For foreign residents, people with speech anxiety, or users with hearing impairments, making such calls remains a significant barrier.

We present **WIGVO**, a server-side relay that bridges a web client and a standard PSTN phone number through two concurrent LLM-backed streaming sessions. The caller speaks (or types) via a browser; the callee answers on an ordinary phone—no app installation, carrier integration, or special hardware is required on either end. WIGVO translates bidirectionally in real time, supporting voice-to-voice (V2V), text-to-voice (T2V), and full-agent modes.

A key challenge is what we call an **echo-induced self-reinforcing translation loop**: TTS audio played to the callee echoes back through the PSTN, is re-recognized as new speech, and gets translated again—creating a runaway feedback cycle. Our solution operates at the architectural level: (i) strict directional separation via two independent Realtime sessions, (ii) deterministic silence injection during TTS playback windows, and (iii) energy-based gating calibrated for μ-law dynamics.

We evaluate WIGVO on 96 completed PSTN calls and report latency, echo suppression, and cost metrics. Our contributions are:

- We formalize the *echo-induced translation loop* problem in streaming speech translation over telephony.
- We propose a *dual-session gated architecture* combining directional separation, silence injection, and energy-based gating for PSTN environments.

System	PSTN	Bidir.	S2S	Echo	A11y
SeamlessM4T	×	✓	✓	N/A	×
Moshi / Hibiki	×	×	✓	N/A	×
Google Duplex	✓	×	×	N/D	×
FCC TRS	✓	✓	×	Human	✓
sign.mt	×	✓	✓	N/A	✓
Vapi / Bland	✓	×	×	N/D	×
WIGVO	✓	✓	✓	✓	✓

Table 1: Comparison with existing systems. PSTN: works over telephone network; Bidir.: bidirectional translation; S2S: automated speech-to-speech (not human-mediated); Echo: handles telephony echo; A11y: accessibility modes. N/D: not disclosed.

- We deploy and evaluate a working relay server across three communication modes, achieving zero echo-induced loops across 96 real PSTN calls with 562 ms median caller-to-callee latency at USD 0.42 per minute.

2 Related Work

Simultaneous speech translation systems such as SeamlessM4T (Seamless Communication et al., 2023b), Seamless Streaming (Seamless Communication et al., 2023a), and ESPnet-ST-v2 (Yan et al., 2023) achieve real-time quality but assume wideband audio from controlled environments. End-to-end full-duplex models—Moshi (Défossez et al., 2024) (200 ms latency), PersonaPlex (Roy et al., 2026) (70 ms turn-taking), and Hibiki (Labiausse et al., 2025) (near-human-interpreter French→English)—all require clean 24 kHz audio and do not address G.711 codec artifacts, telephony echo, or narrowband VAD failures. Hibiki's limitation to a single language pair underscores the difficulty of scaling end-to-end S2ST; WIGVO's modular cascade enables language extensibility through component substitution. In telephony AI, Google Duplex (Leviathan and Matias, 2018) performs monolingual task completion over PSTN, while Vapi[1] and Bland.ai[2] provide LLM-powered phone agents without cross-lingual translation. For accessibility, FCC Telecommunications Relay Services (Federal Communications Commission, 2024) require human intermediaries limited to specific countries, while sign.mt (Moryossef, 2024) targets sign language rather than telephony. Table 1 positions WIGVO against these systems.

[1] https://vapi.ai
[2] https://bland.ai

3 System Architecture and Echo Gating

Figure 1 shows the high-level architecture. A browser client connects to the relay server via WebSocket, sending 16 kHz PCM audio. The relay manages two independent Realtime LLM sessions and a telephony gateway. An `AudioRouter` dispatches events to one of three pipelines—V2V (streaming ASR+TTS in both directions), T2V (typed text to TTS for the callee, speech-to-captions for the caller), and Full-Agent (autonomous call management)—via the Strategy pattern.

Dual Realtime sessions. Session A receives browser audio (PCM16) and produces translated G.711 μ-law for the telephony gateway. Session B receives PSTN audio (G.711 μ-law, 8 kHz) and produces translated audio and captions for the browser. Each session maintains its own system prompt, sliding context window (6 turns), and event stream, ensuring strict directional separation.

3.1 The Echo Loop Problem

In a naive single-session design, TTS output played to the callee echoes back through the PSTN after 80–600 ms, gets re-recognized, and triggers another translation cycle—generating progressively distorted paraphrases until manually interrupted. A Pearson-correlation echo detector comparing outgoing TTS with incoming audio failed in production due to μ-law nonlinear quantization and variable delays. Without gating, 8 of 10 test calls produced echo loops; the correlation detector reduced this to 3 of 10 but introduced frequent false positives. Echo Gate eliminated loops entirely (0 of 96 production calls).

3.2 Echo Gate and 3-Stage Pipeline

PSTN audio entering Session B passes through a 3-stage filter pipeline (Figure 2):

Stage 0: Echo Gate. When Session A begins streaming TTS, an *echo window* is activated (duration estimated from byte length, capped at 1.2 s). Incoming PSTN audio is replaced with μ-law silence (0xFF) before forwarding to Session B. A 0.3 s dynamic cooldown (0.5 s for T2V) accounts for PSTN jitter, followed by a 2.0 s post-echo settling window that suppresses AGC recovery noise while permitting high-energy breakthrough ($\geq$400 RMS) for genuine callee speech.

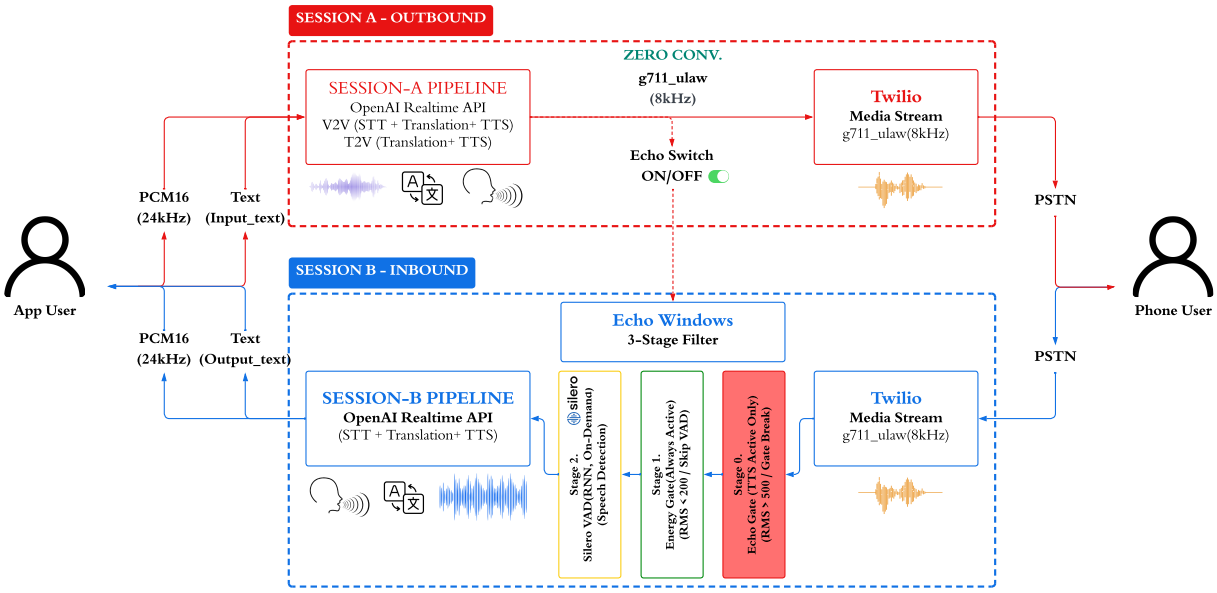

Figure 1: WIGVO system architecture. Session A (red) translates user speech and outputs G.711 directly to Twilio; Session B (blue) receives PSTN audio through a 3-stage filter pipeline (Echo Gate → Energy Gate → Silero VAD). The Echo Gate (dashed) prevents feedback loops by injecting μ-law silence during TTS playback.

Silence injection addresses a distinct failure mode beyond comfort noise generation (Zopf, 2002): if the relay simply *drops* audio during echo windows, the Realtime API's server-side VAD never observes a silence-to-speech transition and remains stuck in a "speaking" state indefinitely. To our knowledge, this failure mode and its mitigation have not been reported in neural streaming S2ST pipelines.

Stage 1: RMS Energy Gate. During echo windows, only signals ≥400 RMS break through as genuine speech; typical echo energy (100–400 RMS) remains suppressed. Outside echo windows, a 150 RMS threshold filters line noise. A three-level interrupt priority (callee > caller > AI TTS) enables natural turn-taking.

Stage 2: Silero VAD. Frames passing the energy gate are processed by Silero VAD (Silero Team, 2021). PSTN audio (8 kHz) is upsampled to 16 kHz via zero-order hold before inference. Asymmetric hysteresis—onset at 96 ms (probability ≥ 0.5, 3 frames), offset at 480 ms (probability < 0.35, 15 frames)—reduces `speech_stopped` latency from 15–72 s (server VAD on PSTN) to 480 ms.

3.3 PSTN VAD Failures and Robustness

The Realtime API's server-side VAD works well for wideband WebRTC audio but fails on PSTN in three ways: (1) constant low-level noise (line hum, codec artifacts) is classified as speech, causing stuck durations of 15–72 s; (2) the noisy PSTN floor prevents energy from dropping below the silence threshold, delaying end-of-utterance detection; (3) audio discontinuities during echo suppression prevent clean silence detection. The 3-stage pipeline addresses all three: the echo gate eliminates discontinuities, the energy gate removes sub-speech noise, and Silero VAD provides neural speech detection with hysteresis. A 15-pattern STT hallucination blocklist intercepts broadcast-style Whisper (Radford et al., 2023) artifacts; 5 hallucinations were blocked across 37 instrumented calls.

Across 96 completed calls, the echo gate activated 127 times (3.4/call) with zero echo-induced translation loops and zero breakthrough incidents.

4 Evaluation

We evaluated WIGVO on 96 PSTN calls (37 fully instrumented) for Korean↔English translation. The remaining 59 calls completed successfully but lacked full per-turn instrumentation due to incremental deployment of the metrics collection pipeline.

4.1 Latency

Table 2 reports latency statistics across the 37 instrumented calls.

Session A achieves 562 ms median latency, within the range for interactive communication.

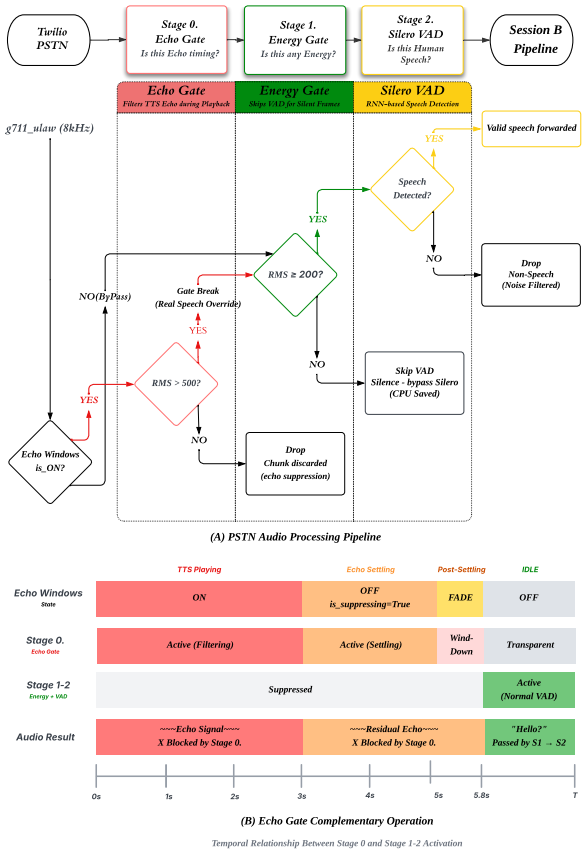

Figure 2: PSTN audio processing pipeline. (a) Three-stage filter between Twilio input and Session B. (b) Temporal behavior: during TTS playback, the Echo Gate replaces audio with μ-law silence while Stages 1–2 are suppressed; after settling, Stage 0 becomes transparent and Stages 1–2 filter PSTN noise.

Session B, which depends on PSTN audio quality and streaming ASR, shows 1,994 ms median. STT (Whisper) accounts for 74.7% of Session B mean latency across 103 paired turns, with translation contributing the remainder. Figure 3 shows the end-to-end latency distributions for Session A and Session B. Figure 4 shows the correlation between utterance length and Session B latency ($r = 0.529$, $p < 0.001$). The five highest Session B outliers (5,981–10,503 ms) all correspond to long callee utterances of 92–125 characters, where streaming STT accounts for ≥95% of E2E latency; the translation step in these cases remains under 1,500 ms. This confirms that tail latency is driven by utterance length and ASR processing time, not architectural overhead.

T2V mode shows lower Session A latency (no caller-side ASR), while Agent mode shows higher

Stage	Avg	P50	P95	Max	N
Session A (ms)	617	562	1,023	2,256	141
Session B E2E (ms)	2,249	1,994	5,390	10,503	207
STT (ms)	2,243	1,969	5,116	—	189
Translation (ms)	763	457	2,255	—	103
First message (ms)	2,585	1,550	7,460	11,308	31

Table 2: Translation latency across 37 instrumented calls. Session A: caller→callee (ASR+translate+TTS). Session B: callee→caller, decomposed into STT (Whisper) and translation (GPT-4o) components. N varies across rows because each metric has a different measurement scope: direction-specific turn counts, successful STT completions, and paired STT+E2E availability for translation decomposition.

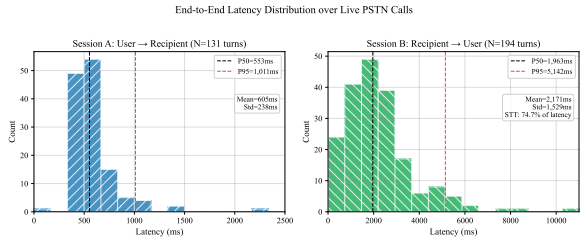

Figure 3: End-to-end latency distributions for Session A (caller→callee, N=141 turns) and Session B (callee→caller, N=207 turns) over live PSTN calls.

Session B latency due to function-calling overhead.

While Session B's ∼2 s median exceeds ITU-T G.114's 150 ms one-way threshold for conversational speech (ITU-T, 2003), that standard applies to same-language dialogue, not cross-lingual translation where additional processing is inherent. A more appropriate baseline is professional simultaneous interpretation, where ear-voice span—the delay between hearing source speech and producing the target—ranges from 2 to 5 s (Goldman-Eisler, 1972; Timaróvá et al., 2011). WIGVO's Session B latency falls at the lower bound of this range. Moreover, the latency is structurally asymmetric by design: Session A (562 ms) keeps the caller-to-callee direction within interactive thresholds, while Session B prioritizes STT accuracy on degraded G.711 audio over speed. Decomposing Session B, STT accounts for 74.7% of mean latency; the translation step itself is 457 ms at P50. This indicates that the bottleneck is narrowband ASR, not the relay architecture. Replacing Whisper with a streaming STT model (Kyutai, 2025) could reduce Session B below 800 ms; the PSTN relay, echo gate, and VAD layers remain necessary regardless.

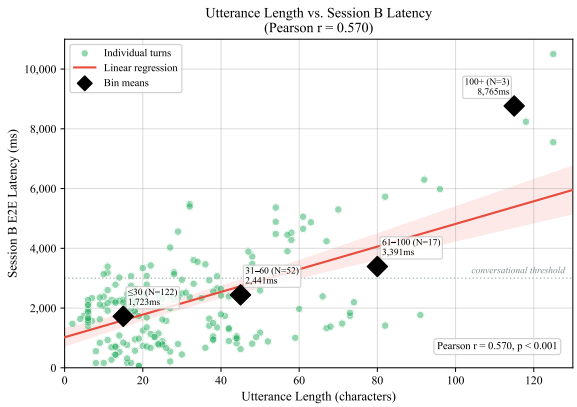

Figure 4: Utterance length (characters) vs. Session B E2E latency. Pearson $r = 0.529$ ($p < 0.001$), indicating that longer recipient utterances incur higher ASR-dominated latency.

Metric	Total	Per-call avg
Echo gate activations	127	3.4
Echo-induced loops	0	0.0
VAD false triggers	59	1.6
Hallucinations blocked	5	0.13

Table 3: Echo, VAD, and guardrail statistics over 37 instrumented calls. Zero echo loops were observed.

4.2 Echo and Safety

Table 3 reports echo suppression, VAD, and guardrail statistics across the 37 instrumented calls.

The echo gate activated 3.4 times per call on average, corresponding to TTS playback events. Critically, **zero echo-induced translation loops** were observed, whereas early prototypes without gating reliably produced such loops. VAD false triggers (1.6/call) were mitigated by a minimum speech duration filter (480 ms hysteresis). The guardrail module blocked 5 hallucinated outputs (broadcast-style phrases from Whisper on low-energy input).

4.3 Cost

Across the 37 instrumented calls (45.5 minutes total), WIGVO consumed 638,035 tokens at a total cost of USD 19.05, yielding **USD 0.42/min** (USD 0.50/call). These costs reflect dual-session Realtime API pricing and remain an order of magnitude cheaper than professional telephone interpretation services (USD 1–3/min).[3] Self-hosted STT alternatives could further reduce per-minute cost; a formal user study with target populations (Brooke, 1996) is planned as future work.

[3]Based on published per-minute rates for over-the-phone interpretation (OPI) from major commercial providers.

Direction	N	BLEU	chrF2++	COMET
en→ko	73	4.1	15.2	0.6883
ko→en	206	9.5	23.6	0.5805

Table 4: Translation quality against Gemini 2.5 Flash offline references. COMET (`wmt22-comet-da`) is a neural metric trained on human judgments; higher is better (max 1.0).

4.4 Translation Quality

We evaluated translation quality by comparing WIGVO's real-time outputs against independent offline references generated by Gemini 2.5 Flash, a separate model family from the GPT-4o Realtime API used for live translation. This cross-model design avoids same-model evaluation bias. From 100 calls with bilingual transcripts, we extracted 73 caller-to-callee (en→ko) and 225 callee-to-caller (ko→en) segments. A noise filter removed 19 callee-side segments (8.4%) containing non-linguistic inputs (consonant-only keyboard artifacts, repeated onomatopoeia, background TV transcriptions).

Table 4 reports BLEU (Papineni et al., 2002), chrF2++ (Popović, 2015), and COMET (Rei et al., 2020) scores computed with SacreBLEU (Post, 2018) and `wmt22-comet-da`.

Surface-level n-gram metrics (BLEU, chrF2++) are low because real-time translation inherently produces valid paraphrases that differ lexically from offline references—a known limitation of these metrics for simultaneous interpretation (Seamless Communication et al., 2023a). COMET, which evaluates semantic adequacy via cross-lingual embeddings, provides a more meaningful signal: en→ko achieves 0.69, indicating adequate translation quality despite surface divergence. The lower ko→en score (0.58) reflects STT errors on narrowband PSTN audio propagating through the pipeline, consistent with the ASR bottleneck identified in the latency analysis above.

4.5 Baseline Comparison

Table 5 contextualizes WIGVO's latency against related systems. Direct comparison is inherently limited because no existing system performs bidirectional cross-lingual speech translation over PSTN: research systems operate in wideband settings with incompatible latency metrics

System	Network	Bidir.	Latency
Seamless Str.	Wideband	×	<2 s AL
Moshi	Wideband	×†	∼200 ms
Hibiki	Wideband	×	2.8–5.6 s LAAL
Google Duplex	PSTN	×‡	N/D
Vapi	WebRTC/PSTN	×‡	∼965 ms*
Bland.ai	PSTN	×‡	<400 ms*
WIGVO SA	**PSTN**	✓	**562 ms P50**
WIGVO SB	**PSTN**	✓	**1,994 ms P50**

Table 5: Latency comparison with related systems. AL: Average Lagging; LAAL: Length-Adaptive AL. †: full-duplex but monolingual (no translation). ‡: monolingual voice agent (no cross-lingual). ∗: vendor-reported; independent reviews report higher values. N/D: not disclosed. Bidir.: bidirectional cross-lingual translation.

(Average Lagging, frame-level), while commercial telephony platforms are monolingual voice agents. We therefore report absolute metrics, as fair head-to-head comparison would require re-implementing competitors in the PSTN environment.

Research S2ST systems report latency on wideband (16–24 kHz) benchmarks. Hibiki (Labiausse et al., 2025) achieves 2.8 s LAAL on short-form French→English but is limited to a single language pair; Seamless Streaming (Seamless Communication et al., 2023a) covers 36 target languages at <2 s AL but does not address telephony. Moshi (Défossez et al., 2024) achieves 200 ms full-duplex latency but performs monolingual dialogue, not translation. None of these systems handle G.711 μ-law, PSTN echo, or narrowband VAD.

Commercial telephony platforms (Vapi, Bland.ai) operate over PSTN but provide monolingual voice agents without translation. Twilio's industry benchmark for AI voice agents targets 1,115 ms median mouth-to-ear turn gap (Twilio, 2025). WIGVO's Session A (562 ms) falls well within this target. Session B (1,994 ms) exceeds it, but this direction includes an additional cross-lingual translation step that monolingual agents do not perform.

5 Demonstration

WIGVO is deployed at https://wigvo-web-gzjzn35jyq-du.a.run.app with a live demo accessible to reviewers.[4] The source code is publicly available at https://github.com/wigtn/wigvo-v2. The web interface supports scenario selection (restaurant, hospital, government office), real-time caption display, and mode switching. Figure 5 shows the interface during an active V2V call.

Figure 5: WIGVO web interface during a V2V call. Left: chat and caption view with bidirectional translations. Right: call status, duration, and mode indicator.

We demonstrate two scenarios:

Restaurant reservation (V2V). An English-speaking user calls a Korean restaurant. Session A translates caller utterances into Korean TTS for the callee; Session B returns Korean responses as English audio and captions.

Medical appointment (T2V). A Korean user with speech anxiety types messages that WIGVO speaks to a clinic via TTS; the clinic's spoken responses appear as real-time captions. A typing indicator is relayed to the callee during text composition.

Beyond translation, the dual-session gated architecture generalizes to multilingual call centers, voice inclusion services, and telephony-native LLM agents combining task completion with cross-lingual communication.

Ethics Statement

All calls were initiated by the authors and collaborators for system testing. Audio was logged for debugging and aggregate metrics only, with no personally identifiable information retained beyond temporary phone numbers. For deployment with real users, informed consent, data minimization, and compliance with privacy regulations (e.g., GDPR, PIPA) are required. The accessibility use case (T2V for speech anxiety) aims to reduce communication barriers but should be deployed with user agency and opt-in participation. The current evaluation covers Korean–English only; the modular cascade architecture supports multilingual ex-

[4] Demo video: https://youtu.be/PLACEHOLDER

tension through component substitution without architectural changes.

Acknowledgments

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